

Quality Assurance and Implementation

CSE 4407

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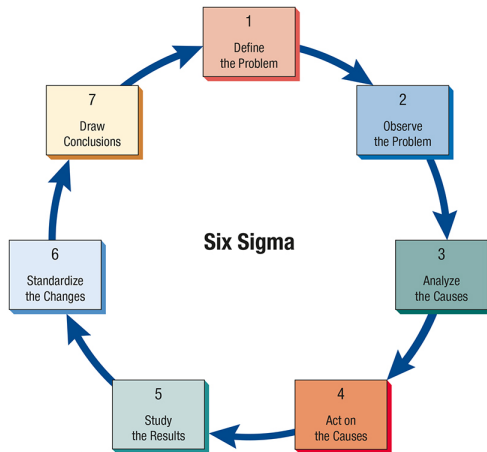


Total Quality Management (TQM)

- **Core Idea:** An evolutionary process toward perfection
- **Shift in Thinking**
 - **Old Way:** Controlling the number of *defective products*
 - **TQM Way:** Building quality into the *entire process*
- **Organizational Context [1]**
 - **Customer** Focus
 - Strategic Planning and **Leadership**
 - Continuous **Improvement**
 - **Empowerment** and Teamwork
- **Goal:** Change behavior and the organization's course, not just fix bugs

Enter Six Sigma: The Quest for Perfection

- **Philosophy:** A culture built on quality, aiming to eliminate *all* defects
- **Goal:** No more than 3.4 defects per 1,000,000 opportunities
- **Approach:** Top-down, requiring CEO adoption and executive champions [2]
- **Methodology:** Continuous cycle of improvement



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Who is Responsible for Quality?

- **Team Sport:** Quality is NOT just the developer's problem
- **Management's Role**
 - Provide full organizational support, not just lip service
 - Establish a context where quality matters
 - Provide time for quality initiatives
- **User's Role**
 - Early commitment to defining quality
 - Help develop and crystallize quality standards
 - Provide feedback

Structured Walkthrough: A Peer-Review Power Tool

- **What:** Peer-review process to monitor development and identify problems early
- **Who**
 - **Author:** The programmer or analyst being reviewed → Listen
 - **Coordinator:** Manages the meeting → Lead
 - **Peer Reviewer:** Points to potential errors → Critique constructively
 - **Notetaker:** Records suggestions → Document
- **Key Rule**
 - The author does NOT defend or argue
 - The goal is to find problems, not to fix them in the meeting
- **Frequency:** Short (30-60 mins) and used routinely throughout the SDLC

Demo: [Walkthrough Report Form](#)

Top-Down vs. Bottom-Up Design

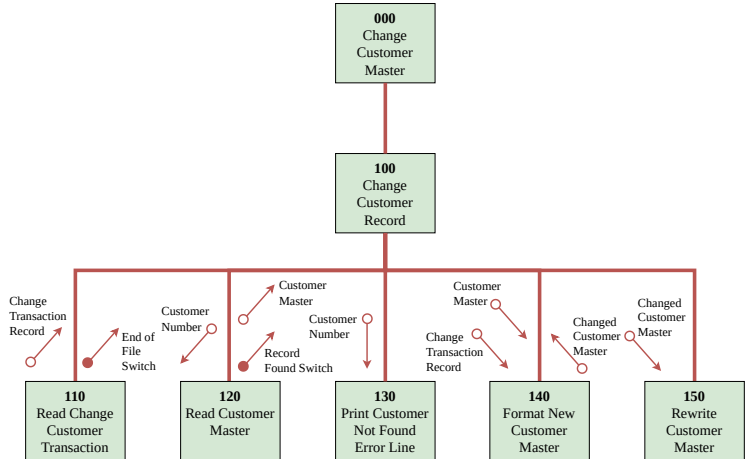
- **Bottom-Up Approach:** The Patchwork Method
 - Build or buy individual department systems first
 - Try to stitch them together later
 - **Cons**
 - Interface bugs
 - Duplicated effort/data
 - Missed overall business goals
- **Top-Down Approach:** The Blueprint Method
 - Start with overall organization objectives
 - Break the system into logical subsystems
 - Design the interfaces first
 - **Pros**
 - Avoid chaos
 - Enable parallel development
 - Keep the big picture in focus

Modular Design with Structure Charts

- **Modular Approach:** Breaking programming into logical, manageable, self-contained portions
- **Why**
 - **Debugging:** Problems are isolated
 - **Maintenance:** Changes are limited to a few modules
 - **Understanding:** Self-contained functions are simple to understand
- **The Tool:** *Structure charts* to visualize the modules and their connections

Example: Structure Chart

- **Rectangles:** Represent the modules
- **Connecting Lines:** Show the hierarchy and calls
- **Data Couples (Open Circle):** Data being passed between modules
- **Control Flags/Switches (Filled Circle):** Signals or status messages being passed

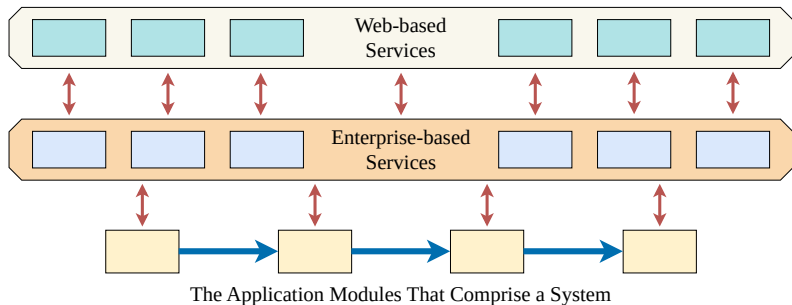


Modern Modularity: Service-Oriented Architecture (SOA)

- **Difference**
 - Loosely-coupled, reusable “services”
 - Not hierarchical like a structure chart
- **Concept**
 - Collection of independent services that can be called upon to perform an action
 - **Examples:** CalculateDaysInMonth(), IsLeapYear(), ReserveHotelRoom()
- **Orchestration:** Connecting these services in a useful way to build an application

How SOA Works

- Services are categorized and can be called by any application module
- Can be general web-based services or specific enterprise-based services



- **Requirements for a Service**
 - Reusable, Interoperable, Modular
 - Categorized, Identifiable, Monitored
- **Challenges:** Industry standards, service management, and security

The Art of Documentation

- **Core Purpose:** Allows stakeholders to “see” the system without interacting with it
- **Why**
 - **Maintenance:** You cannot fix what you do not understand
 - **High Staff Turnover:** Shortens the learning curve for new team members
 - **Quality:** A well-documented system is an easier-to-improve system
- **Documentation Dilemma**
 - **Why does it get skipped:** Lack of time, no rewards, fear of documenting someone else's work
 - In Agile, “It is ultimately the software creator who is responsible for documentation” [3]

Procedure Manuals: The Official Story

- **What:** The English-language guide to the system for users
- **Contents**
 - Introduction and “How To” guides
 - Troubleshooting
 - Technical reference
 - Index and Contact info
- **Complaints**
 - Poorly organized
 - Hard to find information
 - User’s specific problem is not covered
 - Not written in plain English

The FOLKLORE Method: The Unofficial Story

- **What:** Technique for gathering system information that is shared among users but rarely written down [4]
- **Goal:** Capture the “tribal knowledge” of the user base
- **How:** Interview users, investigate files, and observe work to collect:
 - Customs
 - Tales
 - Sayings
 - Art Forms
- **Advantage:** Structured and proactive

The Four Categories of FOLKLORE

- Customs
 - How users currently get the system to work
 - E.g., “We always reboot the server at 11 PM before running the monthly report”
- Tales
 - Stories about problems, effects, and solutions
 - E.g., “That one time Bob from accounting deleted the master file, and here’s how we got it back...”
- Sayings
 - Brief, memorable pieces of advice or generalizations
 - E.g., “Never run the batch process without a fresh backup.”
- Art-Forms
 - User-created diagrams, flowcharts, or tables
 - E.g., a cheat sheet with error codes taped to a user’s monitor

Choosing Your Documentation Technique

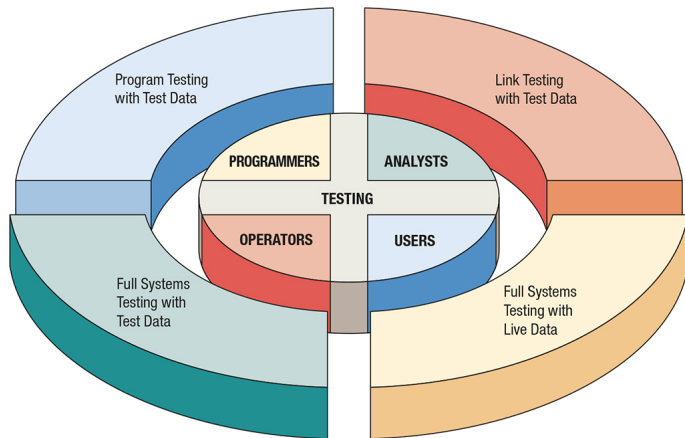
- Choice depends on the context
- Decision Points
 - **Compatibility** with existing documentation
 - **Understood** by others
 - Allows you to **return** to the project after time away
 - **Suitability** for the system's size and complexity
 - Supports a **structured** approach if needed
 - Allows for **easy modification** and updates

Testing, Maintenance, and Auditing: The Final Mile

- Stages
 - **Testing:** Does the system work as designed?
 - **Maintenance:** How do we keep the system working and relevant over time?
 - **Auditing:** How do we verify the reliability of the system's information?
- **Goal:** Deliver and sustain a high-quality system that meets user needs reliably

The Testing Process: Finding Problems Early

- **Mindset**
 - Find previously unknown problems
 - Not to prove perfection
- **When**
 - Continuously
 - Done on modules, subsystems, and the full system
- **Who:** Different people have different roles in ensuring quality



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Level 1: Program and Link Testing

- **Program Testing:** The Micro Level
 - Performed by programmers
 - **Desk Checking:** Manually step through code on paper to find logic errors
 - **Test Data:** Create both valid and invalid data to see if the code handles routines and errors correctly
- **Link Testing:** String Testing
 - Performed by analysts
 - Check if interdependent programs or modules work together as planned
 - Uses a wider variety of test data to simulate normal transactions and error conditions across modules

Level 2: Full Systems Testing

- **Full Systems Testing with Test Data: The Dress Rehearsal**
 - Performed by operators
 - **Checks**
 - Are procedure manuals clear?
 - Do workflows actually flow?
 - Is the output correct and understandable?
- **Full Systems Testing with Live Data: Opening Night**
 - Performed by users
 - Tested with data that has been processed by the old system
 - Allows for direct comparison of output
 - **Crucial Step**
 - Observe real users interacting with the system
 - Listen to their feedback and watch for their reactions

Maintenance: The System's Long Life

- **Reality:** Software maintenance can consume over 50% of an IT budget
- **Cause:** ~70% of software errors are attributed to poor design
- **Goal:** Design flexible, adaptable systems to reduce future maintenance costs
- **Types**
 - **Improvement:** Enhancing the software
 - **Adaptive:** Responding to changes in the organization
 - **Emergency:** Responding to a crisis or failure
- **Key:** Ensure clear user feedback channels are in place

Auditing: The Expert's Review

- **What:** Having an expert, not involved in the project, examine the system to ascertain the reliability of its information
- **Types**
 - **Internal Auditors**
 - Work for the same organization
 - Focus on ensuring internal controls and security are adequate
 - Report to independent management, not the project team
 - **External Auditors**
 - Hired from the outside
 - Typically used when the system impacts financial statements
 - Ensure the fairness and accuracy of financial data
 - May be brought in to investigate issues like fraud

Implementing Distributed Systems

- **Reality:** Systems live on networks
- **Distributed Systems:** Processing is delegated across multiple machines
- **Key Architecture**
 - Client/Server Model
 - Peer-to-Peer Networks
- **Focus:** Logical network design and modeling

The Client/Server Model

- **Concept**
 - Client requests services
 - Server fulfills that request
- **Goal**
 - Feel like a single, unified system
 - Even though processing is distributed
- **Common Architectures**
 - **Two-Tiered**: Client requests directly from a server
 - **Three-Tiered**: An intermediate “application server” sits between the client and the database server to handle business logic

Client/Server Model: Pros, Cons, and Realities

- Advantages

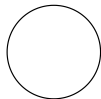
- Greater computer power is available to users
- Greater opportunity to customize applications
- Users are the center of the work

- Disadvantages

- High start-up or switch-over costs are well-documented
- Applications must be written as two separate but cooperative components
- Requires significant changes to organizational culture and work procedures
- Lower processing costs are often claimed but not always proven

Network Modeling: Visualizing the System

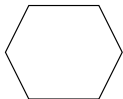
- **Why:** Understand and communicate the logical design of the network
- **Standard Symbols**



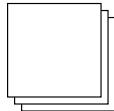
Hub or Local Area
Network



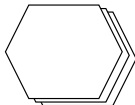
Workstation



External Network



Multiple Workstations
with Similar Functions



Multiple External
Networks with Similar
Functions

Network Modeling Steps

- **Step 1: Decomposition**

- **Purpose:** Provide a high-level overview of the major hubs and workstations in the system
- “Satellite view” of your entire network

- **Step 2: Hub Connectivity**

- **Purpose:** Show how the major hubs are connected to each other and to external networks
- “Country road map” view showing the main highways
- Often includes useful information like distances between hubs

- **Step 3: Workstation Connectivity**

- **Purpose:** Explode a hub and show the detailed connectivity within that hub
- “City street map” view showing every building and local road
- Shows connections between individual workstations and their connections to other hubs

Training Users: The Human Element

- **What:** An educational process to prepare users for a new or modified system
- **Analyst's Role:** Often a manager of training, not just a trainer
- **Most Prized Asset:** Ability to see the system from a new user's viewpoint → Empathy
- **Remember:** Never forget what it feels like to face a new system for the first time

Training Strategy: Who's Who?

- Whom to Train?

- Everyone whose job is affected, from data entry to decision-makers
- Rule
 - Separate novices from experts
 - Training should be tailored to skill level to avoid boredom and confusion

- Who Trains?

- Vendors
 - Experts on their COTS software
 - Often provide focused off-site training
- System Analysts: Know the system and the organization
- External Paid Trainers: Broad teaching experience, but may lack specific context
- In-house Trainers: Know the people and culture, but may lack deep technical expertise
- Other System Users: Cost-effective, but requires good resources for the new trainers

Guidelines for Effective Training

- **Establish Measurable Objectives:** What should trainees be able to do after the session?
- **Use Appropriate Training Methods:** How will you teach them?
- **Select Suitable Training Sites:** Where will the training take place?
- **Employ Understandable Training Materials:** What resources will they have?

Guidelines Deep Dive: Design

- Training Objectives

- Must be clear and measurable
- Lets trainees know what is expected
- Allows for evaluation of training effectiveness
- **Example:** “After training, an operator can successfully troubleshoot the three most common error messages”

- Training Methods

- Match the method to the person and task
- Use a mix of:
 - **Visual Learners:** Demonstrations, manuals
 - **Auditory Learners:** Lectures, Q&A sessions, discussions
 - **Kinesthetic Learners:** Hands-on experience, tutorials, cases

Guidelines Deep Dive: Delivery

- Training Sites

- Off-Site

- **Pros:** Focused environment, no job distractions
 - **Cons:** Away from the real work context, can be costly

- On-Site

- **Pros:** Real-world context, convenient
 - **Cons:** Trainees feel guilty about regular work, constant interruptions

- Training Materials

- Training manuals, cases, and exercises
 - Prototypes and mockups
 - Online tutorials, simulations, and FAQs
 - **Key:** Must be clearly written for the right audience with minimal jargon

Conversion: The 'Go Live' Moment

- **What:** Physically changing from an old information system to a new or modified one
- **Golden Rule:** Right strategy depends on the situation
- **Key to Success**
 - Meticulous **planning** and scheduling
 - Strategic **involvement** of **users**
 - Full file **backup** before you begin
 - Adequate **security** measures in place

Conversion Strategies

Strategy	Description	Pros	Cons	Best For
Direct Changeover	Stop old, start new	Inexpensive, fast	High risk; no fallback	Small systems; when delays are tolerable
Parallel	Run both systems	Safest; allows comparison	Expensive; doubles user workload	Critical systems where errors are unacceptable
Gradual	Slowly ramp up volume	Users adapt slowly; errors are contained	Takes a long time	Agile methodologies; systems with high volume
Modular	Convert one module at a time	Low risk; users learn in chunks	Can be complex to manage dependencies	Object-oriented systems; well-defined modules
Distributed	Convert one location at a time	Problems contained to one site	Each site has unique cultural challenges	Franchises, banks, etc.

Conversion Logistics: The Analyst's Checklist

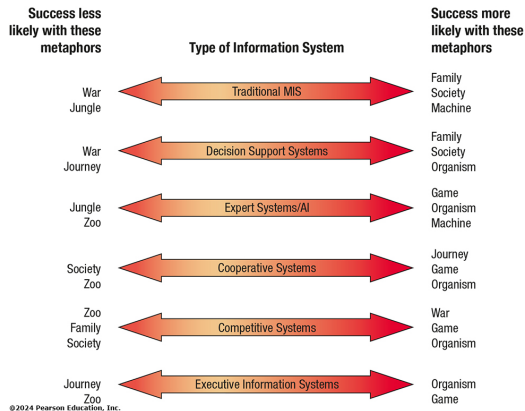
- Key Activities

- **Order Equipment:** Plan for lead times and supply chain issues
- **Order Supplies:** Forms, paper, toner, storage media, etc.
- **Prepare the Site:** Supervise the physical installation
- **Convert Data:** Plan and manage the process of converting all relevant files and databases

- Use project management tools to coordinate these efforts

A Different Lens: Organizational Metaphors

- **What:** Understanding organization's culture by listening to how members describe it
- **Vibe**
 - Zoo or Jungle ↔ Chaos, lack of clear goals, individualistic
 - Machine or Game ↔ Orderly, rule-based, goal-oriented
 - Family or Society ↔ Rules and order, but focused on people, not just goals
 - War or Journey ↔ Goal-oriented, but in a chaotic, competitive environment
 - Organism ↔ A balance between order and chaos, corporate and individual goals



System Security: Building the Fortress

- **Mindset**
 - Not an absolute state
 - Continuum from totally open to totally secure
 - Move systems toward the secure end
- **Shared Responsibility**
 - Only as good as the weakest link
 - Responsibility of everyone who interacts with the system
- **Pillars of Security**
 - Physical
 - Logical
 - Behavioral

The Three Pillars of Security

- Physical Security

- Securing the facility and equipment
- Examples
 - Locked server rooms
 - Biometric access
 - CCTV cameras
 - Off-site backups

- Logical Security

- Securing the software and data
- Examples
 - Passwords
 - Firewalls
 - Encryption
 - Access control lists

- Behavioral Security

- Securing the human element
- Examples
 - Employee screening
 - Written security policies
 - Monitoring logs
 - Security training

A Modern Essential: Two-Factor Authentication (2FA)

- What
 - Two-step verification process
 - Requires users to provide to different authentication factors [5]
- Based on
 - **Something you KNOW:** Password/PIN/Passphrase
 - **Something you HAVE:** Phone/Security token/Badge
 - **Something you ARE:** Fingerprint/Facial scan/Gait
- **Future:** Three-Factor Authentication → Higher confidence (Supposedly), higher inconvenience

Securing E-commerce: Key Precautions

- Threat Detection
 - Malware detection software
 - Intrusion detection and anti-phishing products
 - Vulnerability management to find weak spots
- Access Control
 - Firewalls, gateways, and Virtual Private Networks (VPNs)
 - Email and URL filtering to block spam and malicious sites
- Data Protection
 - Encryption technologies (SSL/TLS, Secure Electronic Transaction)
 - Public Key Infrastructure (PKI) and Digital Certificates to verify senders

The Challenge of IoT Security

- **Internet of Things (IoT):** The network of connected devices like digital assistants, thermostats, sensors, etc.
- **Risk**
 - Rush to connect devices not matched by determination to secure them
 - Easy path for hackers into your network
- **Precautions**
 - **Inventory** all connected devices
 - Ensure all devices are **trustworthy**
 - Identify **risk threats** from vulnerable devices
 - **Mitigate** risks with updates, upgrades, or removal
 - **Monitor** all devices continuously for cyber risks

The Other Side of the Coin: Privacy

- **Trade-off:** More security requires users to give up some privacy
- **Guidelines**
 - **Corporate Policy:** Make it prominent and easy to find
 - **Ask Only What You Need:** Do not collect data if it is not essential
 - **Optional:** Always give customers the choice not to provide personal info
 - **Anonymous Data:** Use profiling technologies that analyze groups, not individuals
 - **Ethics:** Avoid “cheap tricks” like screen scraping or email cookie grabbing

When Disaster Strikes: Recovery Planning

- What
 - Continue operating after a disaster
 - Restore essential IT systems
- Elements of Disaster Recovery Planning
 - Identify the crisis management **teams**
 - Eliminate **single points of failure**
 - Determine data **replication technologies**
 - Create detailed **relocation and transportation plans**
 - Establish multiple **communication channels**
 - Provide **recovery solutions**, including an off-site location
 - Ensure the **physical and psychological well-being** of employees

Disaster Recovery in Practice

- **Data Redundancy**
 - **Data Mirroring:** Nearly real-time backup, but can be affected by distance (>100 miles)
 - **Asynchronous Replication:** Sends data backups at set intervals
 - **Storage as a Service (STaaS):** Renting storage from cloud vendors is becoming more common [6]
- **Communication**
 - Distribute one-page memos with evacuation routes
 - Emergency hotlines or web pages in case email fails
- **People First**
 - Personal and workplace disaster kits
 - Plentiful water
 - Plan for digital conversion of paper documents [7]

So That's It Huh?

- **What:** Assessing the system and network after implementation
- **Why**
 - Feedback for improvement
 - Measure success
- **Common Technique**
 - **Cost-Benefit Analysis:** Does it save money?
 - **Revised Decision Evaluation:** Does it improve decisions?
 - **User Involvement Evaluation:** Were users happy with the process?
- **Comprehensive Method:** The Information System Utility Approach

The Information System Utility Approach

- **Concept:** Measure a system's success by assessing its “utility”
- **Framework:** Evaluate Possession, Form, Place, and Time
 - Who?
 - What?
 - Where?
 - When?
 - How?
 - Why?

The Six Utilities of Information

Utility	Question Answered	Description
Possession	Who should get the info?	Information has no value if the recipient cannot act on it
Form	What does the info look like?	The format, jargon, and calculations must be appropriate for the user, avoiding information overload
Place	Where is the info delivered?	The output must be delivered to the location where the decision is made
Time	When is the info delivered?	It must arrive before the decision, but not so early that it is forgotten or becomes stale
Actualization	How is the info introduced/used?	The system be implementable and maintainable, the users must adopt it
Goal	Why is this info useful?	The system's output must help the organization achieve its strategic objectives

Case Study: Evaluating a Blood Inventory System

- Rule of Utility
 - One “Poor” rating → Module fails
 - All “Good” ratings → Full success
 - Any “Fair” rating → Partial success
- Case Study: Blood Inventory System
- Success Example: Management Summary Reports
 - Rated Good in all 6 utilities
 - Created with user input
 - Delivered info on time, in the right format
 - Helped reduce blood waste
- Failure Example: Heuristic Allocation
 - Users distrusted “mysterious numbers”
 - Too many stakeholders, no buy-in
 - Did not match urgent needs

Evaluating Corporate Websites

- Two-Pronged Approach

- IS Utility Approach

- Checks design, content, and purpose
 - Answers whether the site is useful and goal-aligned
 - Focus on form, possession, and goal of information

- Web Traffic Analysis

- Tracks real user behavior via server data
 - Answers what the visitors are actually doing
 - Moves from assumptions to data-driven insights

Web Traffic Analysis

- How much traffic? → Track hits, sessions, page views
- What content is popular? → Top pages, paths, cart drop-offs
- Who are the visitors? → Demographics, new vs. returning
- Are forms working well? → High error = bad design
- How are users finding us? → Referrals, search engines, keywords
- What tech are they using? → Browsers, devices, versions
- Is advertising effective? → Clicks on campaigns and promos

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